



Development of cracks in soil: An improved physical model

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ABSTRACT

Cracks are common in nature and affect a number of soil properties. They increase the soil permeability, reduce the soil strength and stability, and thus affect the growth of plant roots. Characterization of crack networks in clay soils is needed to quantify the soil-plant relationships. This paper presents a numerical model of the desiccation cracking process, which is based on a lattice of Hookean springs with finite strength and critical strain. A new parameter, i.e., correlation length of critical strain, was used to reflect the spatially-correlated heterogeneity of the soil and to construct a critical strain random field. Three morphological parameters, which are indicators of quantitative crack analysis (i.e., area densities, length densities, and Euler number), were used to describe the spatial distribution and morphological connectivity of cracks. Field experimental results were used to evaluate the numerical model. The results showed that the model replicates the dynamic development of the crack network in nature and the general trends of shrinkage due to drying and crack development observed in the field. In addition, crack formation dynamics and final crack patterns were evaluated during the sensitivity analysis of correlation length. The model's simulation of surface soil cracking was effective in predicting the formation of farmland soil cracks in clay soils.

1. Introduction

The development of cracks in clayey or expansive soils is a common physical phenomenon (Acharya et al., 2015; Sima et al., 2014; Vo et al., 2017). Desiccation cracks can reduce the shear strength of soil and provide a preferential flow path for water infiltration into the soil. Such preferential flow may significantly increase the hydraulic conductivity of soil (Li et al., 2012) and lead to the insufficient treatment of waste contaminants (Flury et al., 1994). Dry shrinkage cracks destroy the integrity of soil and greatly reduce the shear strength and stability of soil. Rainwater may infiltrate into the deeper soil profile along the cracks, consequently reduce the strength of the soil and exacerbating the soil erosion of the slope, which results in failure of the slope (Baker, 1981; Gao et al., 2015; Li et al., 2011). Soil slope instability issues are highly detrimental in a number of engineering applications (Greve et al., 2010; Li et al., 2016). In addition, cracks can lead to dam failures (Peng and Zhang, 2012; Xu and Zhang, 2009), liner failures (Tay et al., 2001), and foundation failures (Lozada et al., 2015). Therefore, researches on crack formation and development can enhance the understanding of a wide spectrum of geologic problems and potentially

benefit numerous geotechnical projects (Li et al., 2012).

Soil desiccation cracking has been studied using both experimental and theoretical approaches. Laboratory experiments have been conducted to investigate the development of desiccation cracks using small-size slurries (Li et al., 2017), clay (Mizuguchi et al., 2005), starch-water mixture cakes (Kodikara and Choi, 2006), and paddy soils (Zhang et al., 2016). However, laboratory experiments may not accurately represent field conditions where the soil often contains coarse particles and different structures, and also has strong interactions with the atmosphere (Wells et al., 2003). Field experiments conducted on in-situ soils under atmospheric conditions would be more realistic for engineering applications (Li and Zhang, 2011).

Desiccation cracking of soil has also been observed in the field, where the characteristic geometries of cracks, such as width, depth, length, thickness, and density, have been investigated under field drying conditions (Gui et al., 2017). When the structure is stationary, the three-dimensional characteristics of the fracture can be inferred using the concept of stereology (Hallaire, 1989; Vogel et al., 1993). The three-dimensional characteristics of cracks can be directly studied by means of geoelectrical measurement (Samouëlian et al., 2003). X-ray

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micro-tomography is a non-invasive way to directly characterize the three-dimensional structure of soil cracks (Vogel and Weller, 2010). In addition, many numerical modeling techniques have been developed to simulate soil cracking (Li and Zhang, 2010; Schanz et al., 2016; Sima et al., 2014). Chertkov et al. (1999, 2000) developed a two-dimensional fracture network model based on probabilistic theory, and described crack width, cross-sectional area, and volume in clay and expansive soil. Horgan and Young (2000) used a discrete element method to simulate the crack network. Péron et al. (2009) used a discrete element method to simulate shrinkage at the microscopic scale, as well as crack initiation, expansion, and geometry. A Spring-Block model was proposed to described the fracturing process (Andersen et al., 1994). A network model based on the subdivision of soil mass in small solid particles connected via springs was recently proposed to simulate the formation of drying cracks in 2-D problems (Musielak and Śliwa, 2012). Maxwell model was proposed to treat clay materials as viscoelastic bodies to simulate the cracking process (Aoki et al., 2002, 2007; Musielak and Śliwa, 2015), and good simulation results were obtained. In addition, molecular dynamics have been used to simulate desiccation cracks in soil (El Youssoufi et al., 2005).

In a previous study, Vogel et al. (2005b) developed a physically based model to simulate crack formation based on Hooke's law. More specifically, their model was used to simulate dynamic crack shrinkage caused by the evaporation of water, where heterogeneity was introduced through the critical threshold of the strain of each spring connecting two nodes. Nodes were identified for each spring with critical strain described by a Gaussian distribution with mean and variance in space. Zhu et al. (2016) later conducted field experiments and used the prior model to simulate crack development. In reality, the soil has a large spatial variability in terms of spatial composition and structure, while the two close-by locations are spatially correlated and not completely independent (Zhang, 2005). Many properties of the soil have been confirmed to have spatial correlations, such as shear strength (Chenari et al., 2019), soil texture (Zhang, 1990), soil hydraulic conductivity (Yeh, 1992). Soil cracking is considered to be related to tensile strength (Corte and Higashi, 1960), elastic properties and moisture content. Thus, it is reasonable to consider that there is a spatial correlation between the properties associated with soil cracking. In nature, the influence of soil roots and plant fibers in farmland causes soil cracking to show obvious spatial continuity or clustering, rather than each location being independent. Therefore, spatial correlation of soil properties should be included during the simulation of soil cracking. Geostatistics is the most common tool to study spatial variability as well as correlation with the help of variogram or correlation function. Li et al. (2019) studied the effect of spatial variability of concrete tensile strength on seismic cracking of gravity dams, and analyzed the sensitivity of correlation length. Although geostatistics theory has been introduced to soil physics and soil engineering problems, it has not yet been applied to the study of soil cracks. Therefore, further improvement of the model proposed by Vogel et al. (2005b) can be achieved.

In this study, we used geostatistical methods to generate a random field for critical strain. In addition to the mean and variance, an additional parameter, correlation length, was introduced to describe the degree of spatial correlation. Then the physically based model of Vogel et al. (2005b) was used to describe the crack formation. Simulation was performed to reproduce the complex pattern of crack development. We use Monte-Carlo simulation to demonstrate the relationship between the model parameter values and crack patterns, and find the optimized parameter set by minimizing the difference between the predicted and measured Minkowski numbers (Mecke, 2000), which can describe the structure of the crack network. The influence of correlation length on the model output is discussed. This approach was used to simulate a series of experiments carried out in farmland.

2. The desiccation cracking model

The desiccation cracking model presented in this paper is an extension and application of the physically based model for crack formation originally developed by Vogel et al. (2005b). A brief introduction of the methodology is as follows. The model constructs a two-dimensional flat layer where the formation of cracks is due to slow contraction of the material as a result of water evaporation. The clay soil layer is regarded as elastic material, which is represented by a triangular network of simple Hookean springs. Each node is connected to its six neighbors by a spring with constant K and natural length d (the length of the relaxed spring). In order to better compare the two models, we chose the same calculation method as Vogel et al. (2005b), so $K = 1$. The strain ε_{ij} of any two nodes, the net force F_i acting on node i through its n neighbors, and the total energy E_i at each node, can be simplified as (Vogel et al., 2005b; Zhu et al., 2016):

$$\varepsilon_{ij} = \frac{|x_i - x_j|}{d} - 1 \quad (1)$$

$$F_i = - \sum_{j \in n} K_{ij} \frac{x_i - x_j}{|x_i - x_j|} [|x_i - x_j| - d] \quad (2)$$

$$E_i = \frac{1}{2} \sum_{j \in n} K_{ij} [|x_i - x_j| - d]^2 \quad (3)$$

where x_i, x_j are the vector positions for any two nodes, $|x_i - x_j| = 1$ is the distance of the nodes in the initial lattice with arbitrary units. And d is the length of the relaxed spring, which is initially set to 1 so that all springs are completely relaxed. Then, desiccation is simulated by decreasing d which leads to an increasing tension of the springs. During desiccation the actual strain and the critical threshold strain are compared, and the spring will break if the strain between two nodes exceeds the critical threshold. The breakage of a spring has immediate consequences for the forces on the nodes previously connected by that spring. They are pulled apart by the other springs, which are still connected. When friction $\mu < F$ the nodes move towards a new location of minimal energy according to Eq. (3). Expansion of the springs $u = 1 - d$ is used as a substitute for time. The boundary condition is specified so that the boundary nodes are not allowed to move perpendicular to the boundary.

According to Vogel et al. (2005b), heterogeneity is introduced through the critical strain threshold of each individual spring connecting two nodes. The threshold for each individual spring is taken from a Gaussian probability distribution with mean and variance, and is distributed randomly as white noise in space, without considering the spatial correlation. Although soil physical and chemical properties always exhibit some degree of spatial variability at various scales due to the heterogeneous nature of geologic formations (Zhu et al., 2018), these properties may be correlated in space and the correlation decreases as the distance between two points increases (Yeh et al., 2015). To represent the soil characteristics described above, a correlation length was introduced into the model to create a spatially-correlated critical strain random field.

The soil critical strain random field was generated by a fast Fourier transform (FFT) algorithm (Cao, 2006; Ravalec et al., 2000). The algorithm produces a random field of real variables based on a Fourier transform of randomized, discrete two-dimensional spectral representations. Assuming the variables have a multivariate normal (or Gaussian) distribution, the spatial persistence is then specified by the auto-covariance function for the variables at two different locations. The purpose of this algorithm is to generate stationary, real-valued processes (Robin et al., 1993):

$$H_i(x) = \int_{-\infty}^{\infty} \exp(2\pi j f \cdot x) dz_i(f) \quad i = 1, 2 \quad (4)$$

where $j = \sqrt{-1}$, x is the node position vector, H is spectral

representation, which is real-valued and stationary. f is the spatial frequency vector, and dz_i is a complex variable, and

$$dz_i = dz_{iR} + jdz_{iI} \quad (5)$$

where subscripts R and I represent the real and imaginary parts, respectively.

Eq. (4) is a Lebesgue-Stieltjes integral, which is used as a guideline to construct realizations with the correct statistical behavior.

The random field is generated according to Eqs. (4) and (5) and the realization is generated with a user-specified random seed number, an isotropic correlation length λ , a mean critical strain value $\bar{\epsilon}$, and a variance of δ^2 . The mean $\bar{\epsilon}$ represents the elasticity of the material, while the variance δ^2 describes its microscopic heterogeneity of soil. Each parameter can be changed to generate different random fields.

To describe the continuity between two points in the random field, a spatial correlation length was introduced to obtain critical strain random fields. To investigate the influence of correlation length on soil cracking, four different values in both x and y directions were employed, i.e., 2.5 cm, 5 cm, 7.5 cm, and 10 cm (The 400×400 node

Table 1

Physical soil properties in the experimental area.

Soil depth/ cm	Particle size distribution (%)			Bulk density (g cm^{-3})	Porosity (%)
	> 50 μm	2~50 μm	< 2 μm		
0~10	3.7	51.0	45.5	1.4253	45.2
10~20	4.6	52.5	45.4	1.5445	44.7
20~30	4.5	51.6	45.3	1.6876	43.8
30~50	4.8	50.4	45.6	1.6424	44.3

represented the $100 \text{ cm} \times 100 \text{ cm}$ area). In the range of isotropic correlation length, the stronger the correlation between two points, the more similar the critical strain between two points. As shown in Fig. 1, we generate four random fields realization with changing correlation length while the other parameters remain constant. As correlation increases, the maximum or minimum values of the critical strain are approximately unchanged, but the threshold distributions are quite different.

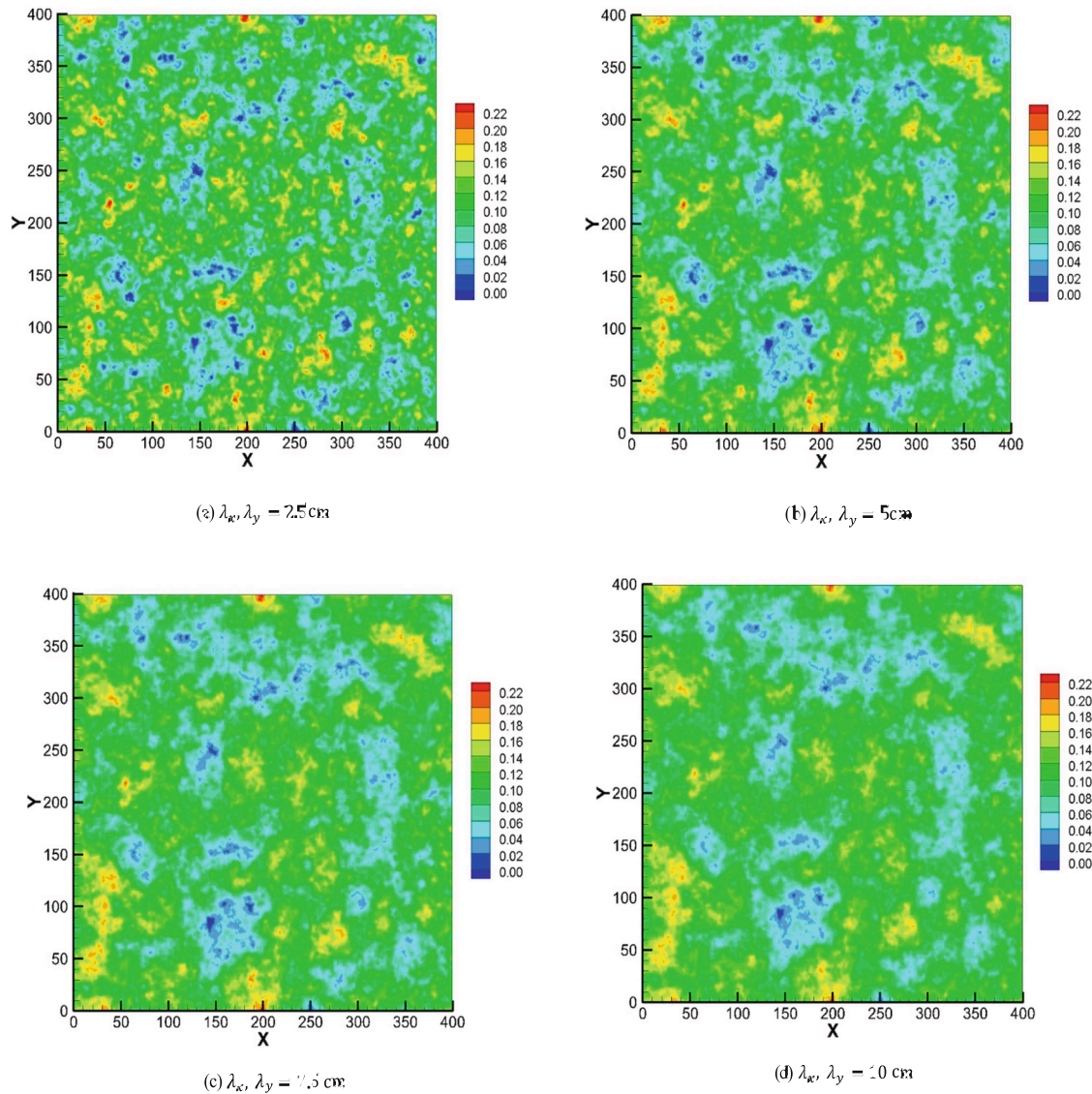


Fig. 1. Critical strain random fields at different correlation lengths. The red area in the picture is of high critical strain while the blue area is of low critical strain. The isotropic correlations are 2.5 cm, 5 cm, 7.5 cm, and 10 cm, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. Materials and methods

3.1. Desiccation cracking experiments in the field

The experiments were performed at the Efficient Use of Water Resources in Arid Modern Agriculture laboratory at the Ministry of Educational Engineering Research Center, in the western part of Ningxia province, China. This area has a temperate monsoon climate with a mean annual temperature of 9.5 °C, average annual precipitation of 202.1 mm, and average annual evaporation of 1947.1 mm. The physical soil properties of the experimental area are shown in Table 1.

The experiments were conducted in six rectangular areas sized 100 cm × 100 cm. Initially, the ponded water was exerted at the soil surface until the infiltration rate stabilized. The measured soil moisture after infiltration ensured that the soil was fully saturated at the upper 10 cm layer. During the process of soil evaporation and cracking, a Canon EOS 5DII digital camera captured images of crack formation and development every two hours with a pixel resolution of 2048 × 2048. The photos were taken during the daytime and a white sheet was placed over each plot to diffuse the light and to avoid directional radiation starting from 8:00 to 18:00 for a continuous period of three days. To avoid the influence of natural rainfall on crack in the experiment area, insulation and rain-proof materials were used to cover the experimental area at night (Fig. 2).

3.2. Photography and image files

To analyze the geometric characteristics (e.g., the width, length, number, and distribution) of the cracks, the digital images needed to be processed. The typical procedure for digital image processing included three steps: 1) the image was applied with geometric correction, and 2) a color image of the crack patterns was converted into a gray image, which was then converted into a binary image using the threshold division method, and 3) the image was processed through noise removal, and crack edge extraction of crack for morphological properties. After this process, the cracks and aggregates were distinguished in a black-white image, where black lines and the white areas represent crack networks and aggregates, respectively. During the threshold division process, the threshold was optimized to make the binary images as clear as possible. If the values of the image pixels were less than the threshold, they were set to zero; otherwise, they were set to one. The “Graythresh” function in MATLAB software was used to obtain the best threshold value for each image and generate the grayscale image.

3.3. Morphological measures

To quantify the morphology of the crack network, three fundamental properties were used to describe the structure of the cracks spatial network: area A , length L , and the Euler number E defining the topological structure of the cracks. These three quantities are known as Minkowski numbers (Mecke, 2000), which can be calculated based on the structure of the binary images (Qiu, 2001; Vogel et al., 2005a). The topology of the image was used to quantitatively describe the characteristics of soil surface cracks (Vogel et al., 2005a).

The area in the calculation is defined as:

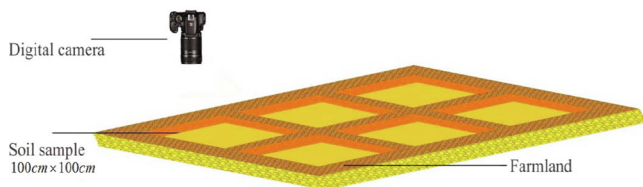


Fig. 2. Experiment layout.

$$M_0(X) = A(X) \quad [L^2] \quad (6)$$

where M_0 is the area of the cracks, that measured by the number of black pixels in the binary images.

The crack length is defined as:

$$M_1(X) = \int_{\delta X} ds = L(X) \quad [L] \quad (7)$$

where M_1 is the boundary length of the crack defined as the integrals over the boundary δX of the structure X . ds is the boundary element. There are many ways to calculate Minkowski length, and Vogel and Weller (2010) proposed an efficient algorithm. In our study, the length of the crack image was calculated using the method proposed by Cao and Wang (2014) with open-source code implemented in MATLAB software. L is the circumference of the isolated crack edge when the crack is an isolated one. On the contrary, L is the grid perimeter when the crack network forms a mesh.

Topological measure describing the connectivity of the structure is defined as:

$$M_2(X) = \int_{\delta X} \frac{1}{r} ds = 2\pi E(X) \quad [-] \quad (8)$$

where M_2 is the topology measured of the crack by the Euler number $E(X)$, which is defined as the number of isolated objects N (closed convex) minus the number of holes C (closed concave), i.e., $E = N - C$. r is the radius of curvature of the boundary, which is positive for convex and negative for concave boundaries. The Euler number of the crack image was calculated using the “Bweuler” function in MATLAB software.

To compare the crack structure from images with different sizes, Minkowski densities are introduced by dividing Minkowski numbers with the area of the image Δ (or $(2\pi\Delta)$):

$$m_c(X) = \frac{M_c}{\Delta} \quad c = 1, 2 \quad (9)$$

$$m_2 = M_2/(2\pi\Delta) \quad (10)$$

where Δ is the size of the image and m_2 is used to describe the density of the Euler number (Vogel et al., 2005a; Zhu et al., 2016).

3.4. Model evaluation method

Model performance was evaluated graphically by comparing the simulated and measured Minkowski densities versus time. In addition, the coefficient of determination R^2 , root mean square error $RMSE$, $BIAS$, and index of agreement (IA) were used as statistical indexes to quantitatively evaluate the difference between the Minkowski densities of the experimental and simulated images. The mathematical expressions for R^2 , $RMSE$, $BIAS$, and IA are as follows (Hao et al., 2015; Leisenring and Moradkhani, 2011):

$$SE = \sum_{i=1}^N (S_{wos,i} - S_{wsm,i})^2 \quad (11)$$

$$SE_{os} = \sum_{i=1}^N (S_{wos,i} - \bar{S}_{wos,i})^2 \quad (12)$$

$$R^2 = 1 - \frac{SE}{SE_{os}} \quad (13)$$

$$RMSE = \sqrt{\frac{SE}{N-1}} \quad (14)$$

$$BIAS = \frac{\sum_{i=1}^N |S_{wos,i} - S_{wsm,i}|}{\sum_{i=1}^N S_{wos,i}} \quad (15)$$

$$IA = 1 - \frac{SE}{\sum_{i=1}^N [|S_{wsm,i} - \bar{S}_{wos,i}| + |S_{wos,i} - \bar{S}_{wos,i}|]^2} \quad (16)$$

In Eqs. (11)–(16): $S_{wsm,i}$ is the simulated value; $S_{wos,i}$ is the observed value; $\bar{S}_{wos,i}$, $\bar{S}_{wsm,i}$ are the average values of $S_{wos,i}$, $S_{wsm,i}$ respectively; and N is the number of samples.

RMSE: Root Mean Square Error; the lower this measure, the better the model's performance. The lower limit, zero, is the optimum value.

IA: The Index of Agreement; a standardized measure of the degree of model prediction error that ranges from 0 and 1. A value of 1 indicates a perfect match, and 0 indicates no agreement.

R^2 : coefficient of determination; a statistical measure of how well the regression predictions approximate the actual data points. An R^2 of 1 indicates that the regression predictions perfectly fit the data.

BIAS: a feature of a statistical technique or of its results whereby the expected value of the results differs from the true underlying quantitative parameter being estimated. The optimum value is 0. Negative values indicate that the model underestimates the observed data, while positive values indicate model overestimation.

4. Results and discussion

4.1. Calibration and validation of the model

Experimental zone 1 was chosen for model calibration and experimental zones 2–6 were used to validate the model. The area, length, and Euler number densities of the experimental and simulated images were computed and compared. In the processes of model calibration and verification, the experimental images and the simulated images were converted to binary images and then noise was removed and crack edges were extracted.

4.1.1. Calibration of the model

The calibration objective was to capture the overall crack pattern observed in the field (Fig. 3(a)) by adjusting the input parameters. We evaluated the improved model and Vogel model parameters using the Monte Carlo method, with a total of 1,481,040 realizations, using twenty-four cores (Inter Xeon Gold 6128 CPUs running at 3.4 GHz with 128 GB of RAM) on a workstation. The parameters are combined within a reasonable range (see Table 2), and the crack pictures simulated by all the parameter combinations are compared with the experimental pictures in terms of Minkowski densities at different time. We intend to find the optimized parameters that produce the smallest simulation error. Please see Table 2 for the details of the model calibration.

Before the formal parameter calibration, the field experimental time t should be matched with the time in the crack simulation model. According to Vogel et al. (2005b), $u = 1 - d$ is used as a substitute for time, where d is the relaxation length of the spring. We tried to minimize the differences between the calculated and simulated Minkowski densities by adjusting the u value (the rest parameter values are given with the medians of their ranges in Table 2) to match the four time points (14 h, 22 h, 44 h, 68 h). The preliminary calibration step have made a match between model time and experimental time, so that the corresponding Minkowski densities at different time can be compared later to formally calibrate the remaining unknown parameters.

We define P_{mi} as well as P according to the three evaluation indexes (i.e., R^2 , IA , $BIAS$) for the purpose of parameter calibration:

$$P_{mi} = R^2 + IA - BIAS \quad i = 0, 1, 2 \quad (17)$$

$$P = \sum_{i=0}^2 P_{mi} \quad (18)$$

where P_{mi} ($i = 0, 1, 2$) is the comprehensive calculated value of the three evaluation indexes of area, length and Euler number densities, respectively. P is the summation of P_{mi} values calculated by area, length and Euler number densities. $P_{mi} \leq 2$ and $P \leq 6$ (since $R^2 \leq 1$, $IA \leq 1$ and $BIAS \geq 0$). The objective of the calibration is then to find an optimized parameter set that can maximize P value.

According to the results of Monte Carlo simulation, the P values of

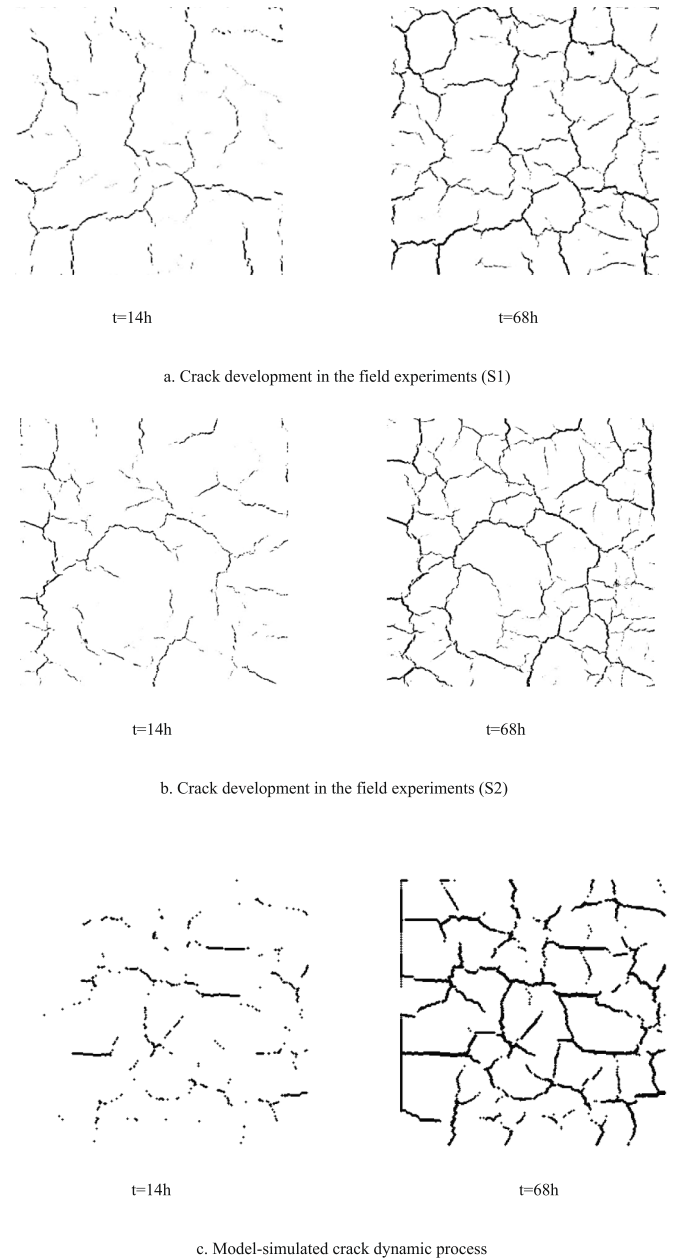


Fig. 3. Comparison of simulated and experimental crack images.

all parameter combinations are calculated to determine the optimal parameters of the model. The RMSE index is also calculated as a reference. The calibrated input parameters for this study and those from Vogel et al. (2005b) are listed in Table 4. It can be seen in Fig. 3 that the simulated crack evolution process was similar to natural crack development. As shown in Fig. 4, a good match between simulated and measured values has been found in the calibration of the model. Table 3 show that the P value of the improved model is 5.7692, and the P value of the Vogel model is 5.156, indicating that the calibration results were satisfactory.

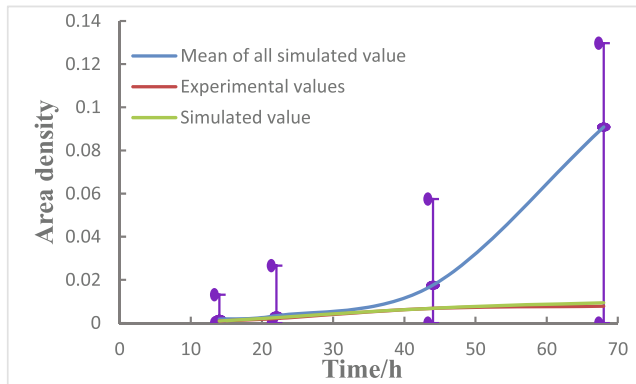
4.1.2. Validation of the model

The validation was conducted by measuring crack image areas at different time in the experimental zones, and 100 soil critical strain random fields were generated using different random seeds for the random field generator (the other parameters are shown in Table 4).

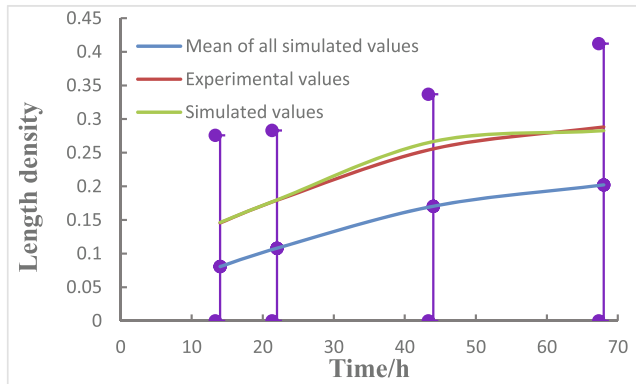
Indices demonstrate the performance of the model validation (Tables 5 and 6). The average IA , $RMSE$, R^2 , and $BIAS$ values were

Table 2
Parameter range and combination of Monte-Carlo approach.

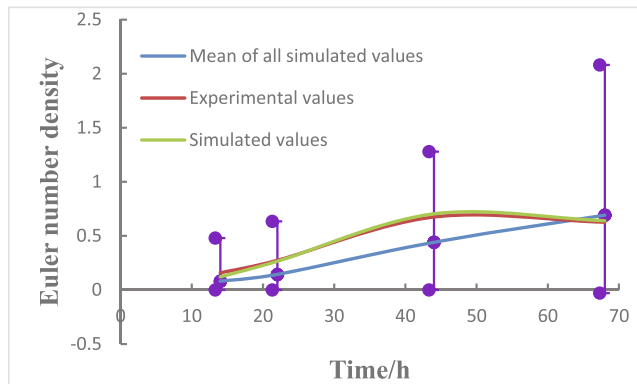
Model parameters	Calibration value in this study		Calibration value of Vogel's model (2005b)	
	range	increment	range	increment
Mean value of critical strain $\bar{\varepsilon}$	0.05~0.15	0.01	0.05~0.15	0.01
Variance δ^2	0.10~0.35	0.005	0.1~0.35	0.005
correlation length λ (cm)	1~10	1	/	/
Friction μ	0.001~0.0015	0.0001	0.001~0.0015	0.0001
Relaxation parameter n_{it}	10~25	1	10~25	1
Shrink minimum length of nodes	0.800~0.999	0.001	0.800~0.999	0.001



(a) Area density



(b) Length density



(c) Euler number density

Fig. 4. Match between simulated and measured values in the calibration of the model. The error bar represents the deviation from the extremum and average of all simulated values.

satisfactory for area density, length density, and Euler number density of the two models. However, the R^2 and IA values in Table 5 were higher than most of the values in Table 6. In the next section the two models are compared.

The mean and standard deviation of Minkowski densities (area, length, and Euler number density) were obtained to further examine the proposed model. The mean and standard deviation of the experiment came from the results of the five experiments, while those of the simulation came from the 100 crack images based on the 100 random field realizations of the critical strain. Fig. 5 displays the temporal change of Minkowski densities and their uncertainties from the improved model, the Vogel model and the experiment, respectively.

4.2. Comparison of the results of the two models

In nature, plant roots can reinforce the soil and inhibit the development of soil cracks (Chen et al., 2017; Cislighi et al., 2017; Zhang et al., 2014). Soil cracks away from the root system are more easily developed than close to the roots. The critical strain of the soil with root is relatively larger due to root reinforcement, while lower value appears at a distance from the root system. The cracking resistance of plant fibers can produce a similar phenomenon (El-Halim, 2017; Tang et al., 2012). These phenomena indicate that it is reasonable to introduce correlation length.

To compare the two models more intuitively, a relative difference variable RD was defined, i.e., $RD = (X_m - X_v) \times 100\% / X_v$, where X_m and X_v are the indexes of the results from the model in this study and from Vogel et al. (2005b), respectively. As indicated in Table 7, the average RD values for R^2 and IA for the area, length, and Euler number density were 26.99% and 8.61%, respectively. In contrast, the average RD values for $RMSE$ and $BIAS$ were -23.20% and -26.72%, which meant that the $RMSE$ and $BIAS$ of the results from current model are smaller than those from the Vogel et al. (2005b) model. This shows that the current simulation results were better than the simulation results from Vogel's model. Therefore, the introduction of correlation scale for a spatially-correlated random critical strain field is more suitable for the simulation of farmland crack development than the original Vogel et al. (2005b) model.

4.3. Effects of correlation length

The model proposed in this paper depended on five parameters: mean critical strain $\bar{\varepsilon}$, variance δ^2 , correlation length λ , friction μ and the relaxation parameter n_{it} . Changing $\bar{\varepsilon}$ does not enrich the variety of crack patterns, while friction μ , the relaxation parameter n_{it} and variance δ^2 have been analyzed in detail in Vogel et al. (2005b). Therefore, we checked the effects of the correlation length λ on the simulation results, while all other parameters remained constant.

The mean value $\bar{\varepsilon}$ was determined to be 0.1, variance δ^2 was 0.33, friction μ was 0.001, and the relaxation parameter n_{it} was 20, while the correlation lengths in both the x and y directions were 2.5 cm, 5 cm, 7.5 cm, and 10 cm.

As shown in Fig. 6, a large number of isolated cracks will occur

Table 3
Calibration statistics for the model area density, length density, and Euler number density.

Statistical indicators	Area density in this study	Length density in this study	Euler number density in this study	Area density of Vogel's model (2005b)	Length density of Vogel's model (2005b)	Euler number density of Vogel's model (2005b)
$R^2[-]$	0.9325	0.9884	0.9893	0.953	0.922	0.856
$IA[-]$	0.9999	0.9996	0.9990	0.882	0.950	0.959
$BIAS[-]$	0.0963	0.0116	0.0316	0.185	0.113	0.068
$RMSE$	0.0011	0.0063	0.0236	0.003	0.003	0.052
P_{m_i}	1.8361	1.9764	1.9567	1.650	1.759	1.747
P	5.7692			5.156		

Note: R^2 , IA and $BIAS$ is relatively index, dimensionless, $RMSE$ is the same as the dimension of the statistical object.

Table 4
Calibrated model parameters for current study and model in Vogel et al. (2005b).

Model parameters	Calibration value in this study	Calibration value of Vogel's model (2005b)
Mean value of critical strain $\bar{\epsilon}$	0.1	0.1
Variance δ^2	0.33	0.125
correlation length λ (cm)	7.5	/
Friction μ	0.001	0.0015
Shrink minimum length of nodes	0.911	0.9
Relaxation parameter n_{it}	20	18

Table 5
Validation statistics for area density, length density, and Euler number density.

Statistical indicators		$R^2[-]$	$RMSE$	$BIAS[-]$	$IA[-]$
Average for S2–S6	m_0	0.877	0.001	0.175	0.991
	m_1	0.776	0.043	0.123	0.987
	m_2	0.869	0.108	0.145	0.984

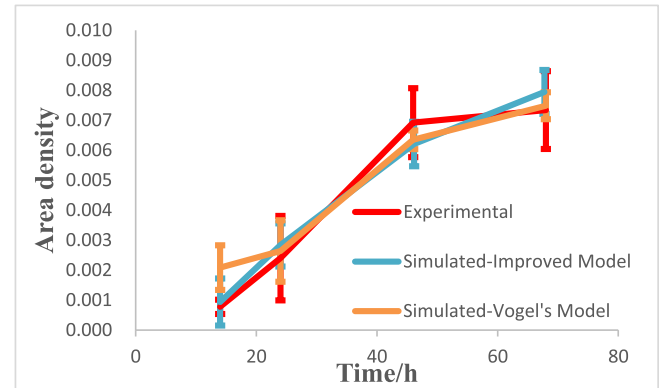
Note: R^2 , IA and $BIAS$ are dimensionless; $RMSE$ has same as the dimension of the statistical variable.

Table 6
Validation statistics for area density, length density, and Euler number density from Vogel's model.

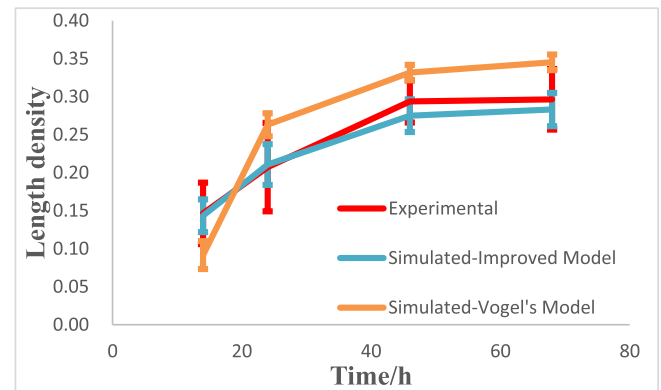
Statistical indicators		$R^2[-]$	$RMSE$	$BIAS[-]$	$IA[-]$
Average for S2–S6	m_0	0.841	0.001	0.222	0.955
	m_1	0.686	0.050	0.161	0.939
	m_2	0.616	0.202	0.264	0.852

when the correlation length is small. As the correlation length increases from 2.5 to 10 cm, the length of a single crack and the area of the aggregate increases significantly. This is due to the similarity of the critical strain of the spring within the range of the correlation length and the linear development of the crack along its tip. This is also a prominent property of natural crack formation. On the other hand, with the increase of the correlation length, there are still small cracks, and the irregularity of the crack pattern still exists, because the correlation length does not affect the overall heterogeneity of the region (Fig. 6(d)).

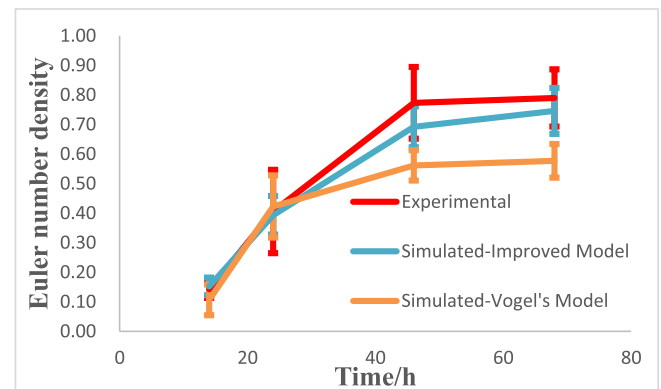
The simulated crack images with different correlation lengths were quantitatively described using the Minkowski densities (Fig. 7). With the evaporation process, the area density and length density increased, reflecting the phenomenon of increasing area and length in the process of crack development. The Euler number density increased first and then decreased, reflecting the connectivity of crack network. The formation of a large number of isolated cracks in the initial stage of the crack network led to the continuous increase of the Euler number density. With the development of cracks, the decreasing Euler number density indicates the formation of crack networks. The variation of the



(a) Area density



(b) Length density



(c) Euler number density

Fig. 5. Comparison between simulated and measured values in the validation of the model. The error bar represents the standard deviations of the simulated and experimental values, respectively.

Table 7

Comparison of the two simulation results for area density, length density, and Euler number density.

RD values	$R^2[-]/\%$	RMSE/%	BIAS[-]/%	IA[-]/%
m_0	4.68	-22.08	-14.98	3.90
m_1	18.57	-9.30	-22.56	5.24
m_2	57.72	-38.23	-42.63	16.68
Average RD	26.99	-23.20	-26.72	8.61

Note: m_0 – m_2 are Average RD values for the five experiments S2–S6.

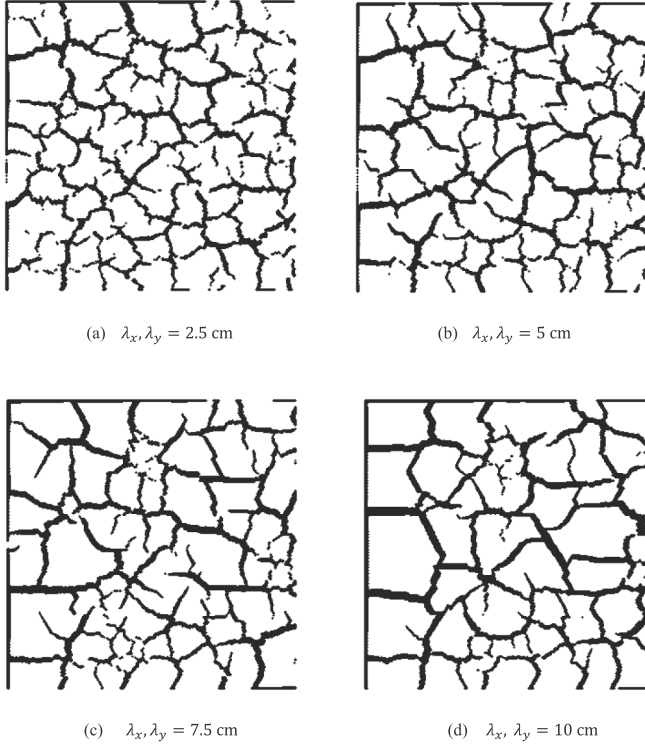
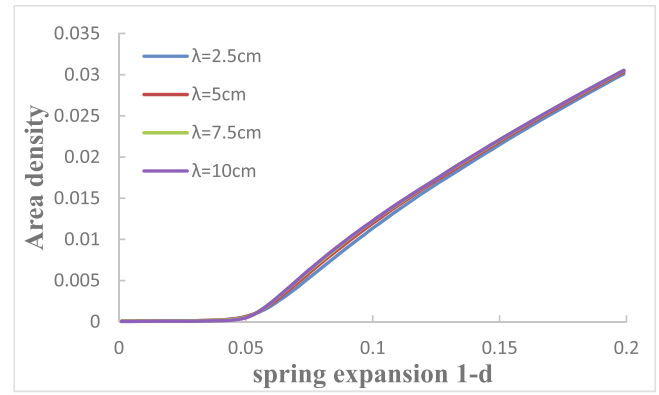


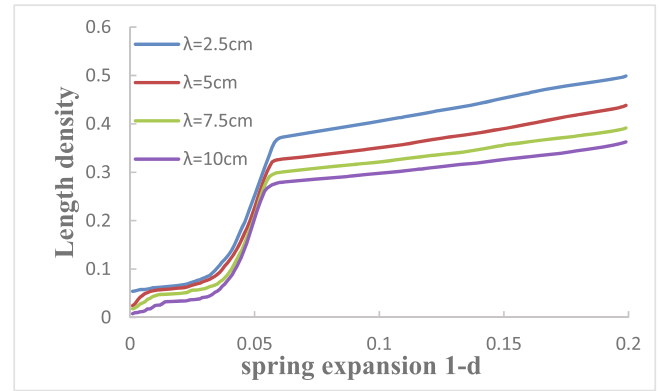
Fig. 6. The length of simulated crack increases with the increase of correlation length ($\bar{\epsilon} = 0.1$, $\delta^2 = 0.33$, $\mu = 0.001$, $n_{it} = 20$).

Minkowski density function with spring contraction in Fig. 7 is similar to those from the Vogel et al. (2005b) model. As shown in Fig. 7(b) and (c), as the correlation length increases, the values of the length and Euler number densities decrease. The springs within the correlation length range have relatively homogeneous critical strain, and the local heterogeneity is weak. The larger the correlation length, the easier it is for isolated cracks to form interconnected crack networks. The decrease in the number of isolated cracks leads to a decrease in the length density and Euler number density (Vogel et al., 2005b). However, the effect of correlation length on area density is not significant (Fig. 7(a)).

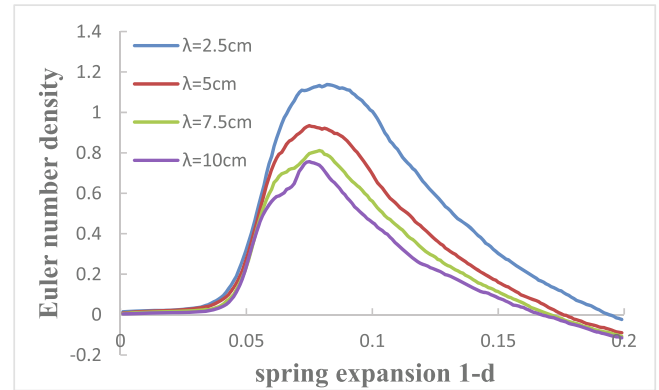
Therefore, the effect of the correlation length and variance on crack pattern and Minkowski density are essential different. The decrease of variance leads to the decrease of heterogeneity, resulting in the relatively regular crack pattern (Vogel et al., 2005b). On the one hand, as the increase of correlation length does not reduce the heterogeneity of the whole region, the crack pattern is irregular and has a large number of small cracks (Fig. 6(d)). The variance change will cause the area density change (Vogel et al., 2005b), but the effect of changes in correlation length on area density is insignificant (Fig. 7(a)). This also indicates that the correlation length changes the distribution of critical strains without affecting the overall heterogeneity of the region. Thus, the effect from changing correlation length on crack pattern cannot be completely compensated by changing variance, which shows the necessity of introducing the parameter of correlation length.



(a) Area density



(b) Length density



(c) Euler number density

Fig. 7. Minkowski function dynamic of simulated crack patterns with increasing correlation length ($\lambda = 2.5$ cm, 5 cm, 7.5 cm, 10 cm).

5. Conclusions

In this paper, a new parameter, correlation length, was introduced to Vogel et al. (2005b) model to reflect the spatial correlation of critical strain during the simulation of crack formation. This new parameter can be optimized using experimental data. The results showed that:

The model, which introduced correlation length, was used to simulate crack development, and the results of the simulated results agreed better with experimental data, compared to results from Vogel's model.

The effect of correlation length on model output was analyzed based on Minkowski density and crack pattern. The larger the correlation

length, the longer the crack length and the larger the aggregate area. The increase of correlation length will result in the decrease of isolated crack, which will result in the decrease of length and Euler number densities. The effect of correlation length on area density is not obvious.

The introduction of correlation length makes the model more suitable for the description of soil dry shrinkage cracking and can better predict the crack development, which can be more widely used in geological engineering and hydrology. For the preferential flow problem, the possible preferential flow path can be predicted more reasonably. Understanding the formation of cracks in engineering geology can provide practical guidance for failure risk analysis and engineering designing.

This model reproduced the dynamic characteristics of crack network development and was able to reproduce the main trends observed in experimental testing (e.g., shrinkage related to drying, and crack development). The simulation reproduced the complex phenomenon of crack pattern development and made full use of geostatistical tools to describe the spatial correlation of soil properties (i.e., critical strain).

Appendix A

Table A1

Validation statistics for area density, length density, and Euler number density.

Statistical indicators		$R^2[-]$	$RMSE$	$BIAS[-]$	$IA[-]$
S2	m_0	0.931	0.001	0.132	0.996
	m_1	0.906	0.036	0.101	0.994
	m_2	0.964	0.069	0.075	0.994
S3	m_0	0.921	0.001	0.157	0.995
	m_1	0.968	0.016	0.048	0.998
	m_2	0.935	0.074	0.107	0.992
S4	m_0	0.856	0.001	0.215	0.988
	m_1	0.810	0.041	0.112	0.987
	m_2	0.902	0.121	0.127	0.987
S5	m_0	0.904	0.001	0.215	0.994
	m_1	0.537	0.064	0.220	0.979
	m_2	0.733	0.150	0.292	0.966
S6	m_0	0.774	0.001	0.154	0.984
	m_1	0.657	0.057	0.136	0.978
	m_2	0.811	0.128	0.117	0.974

Note: R^2 , IA and $BIAS$ is relatively index, dimensionless, $RMSE$ is the same as the dimension of the statistical object.

Table A2

Validation statistics for area density, length density, and Euler number density from Vogel's model.

Statistical indicators		$R^2[-]$	$RMSE$	$BIAS[-]$	$IA[-]$
S2	m_0	0.871	0.001	0.190	0.967
	m_1	0.860	0.041	0.131	0.966
	m_2	0.515	0.254	0.299	0.803
S3	m_0	0.848	0.001	0.249	0.959
	m_1	0.560	0.059	0.176	0.913
	m_2	0.736	0.149	0.235	0.910
S4	m_0	0.941	0.001	0.140	0.982
	m_1	0.797	0.042	0.119	0.959
	m_2	0.340	0.314	0.343	0.718
S5	m_0	0.810	0.001	0.306	0.943
	m_1	0.408	0.065	0.235	0.895
	m_2	0.789	0.134	0.238	0.927
S6	m_0	0.735	0.002	0.224	0.922
	m_1	0.807	0.040	0.143	0.962
	m_2	0.698	0.157	0.207	0.900

Table A3

Comparison of the two simulation results for area density, length density, and Euler number density.

RD values	$R^2[-]/\%$	RMSE/%	BIAS[-]/%	IA[-]/%
S2	6.89	-36.36	-30.52	2.99
	5.35	-12.20	-22.90	2.90
	87.18	-72.84	-74.92	23.79
S3	8.61	-27.27	-36.95	3.75
	72.86	-72.88	-72.73	9.31
	27.04	-50.34	-54.47	9.01
S4	-9.03	-7.69	53.57	0.61
	1.63	-2.38	-5.88	2.92
	165.29	-61.47	-62.97	38.16
S5	11.60	-9.09	-29.74	5.41
	31.62	-1.54	-6.38	9.39
	-7.09	11.94	22.69	4.21
S6	5.31	-30.00	-31.25	6.72
	-18.59	42.50	-4.90	1.66
	16.19	-18.47	-43.48	8.22

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoderma.2020.114258>.

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